

Constrain the Target, Not the Loss: Action-Space Trust Regions for Critic-Guided Policy Fine-Tuning

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Abstract

Behaviour-cloned robot policies are capped by their demonstrations, and the usual way past that cap is to train a critic and fine-tune against it. This reliably goes wrong: the policy drifts away from the data the critic was fit to, its score rises, and its actual success collapses.

We present an actor update that does not. A frozen base proposes a cloud of candidate actions; each is displaced along the critic’s gradient; a restoring force returns any candidate that has strayed beyond a radius ρ of the base; and a small residual head is regressed onto the displaced candidates. Nothing else in the pipeline changes — the critic, replay, demonstrations, budget and evaluation are those of the state-of-the-art residual fine-tuner, byte for byte, so the only variable is the actor update.

Under that control, the update gains +9.4 points on the eight hardest LIBERO-90 tasks where backpropagating $-Q$ gains +0.9, and lifts robomimic square from 0.382 to 0.91–0.93. It reaches the published flow-matching pipeline’s performance from a base less than half as strong, and transfers unchanged to that flow-matching base.

We then ask which part is responsible, switching each guard off independently on two tasks. The answer is decisive in one direction only. Removing both is catastrophic: success falls to exactly 0.000 on both tasks, from bases of 0.573 and 0.610, so the unconstrained action-gradient update — what prior work in this family runs — does not merely underperform here, it destroys the policy. But either guard alone recovers essentially all of the gain (0.671–0.803 across configurations, against 0.781 and 0.684 for both together), and the gaps between them are smaller than this benchmark’s seed-to-seed spread. We therefore claim only that a constraint tying the policy to its base is necessary; which constraint, at this budget, we cannot distinguish.

1 Introduction

A behavior-cloned robot policy is only as good as its demonstrations. To push past them, the usual move is to learn a critic and fine-tune against it with reinforcement learning. This paper is about one design choice inside that loop, which we think is made wrongly almost everywhere: how much authority the critic is given.

A critic does two jobs in a standard update. It says which way the policy should move, and — through the magnitude of its gradient — how far. Only the first is a job it was trained for. A critic learns to rank actions in the region its data covers; outside that region nothing constrains it, and it is quite capable of being steep and wrong at the same time (Fujimoto et al., 2018; 2019; Jain et al., 2025). The consequences of the two errors are not comparable. A slightly wrong direction gives a slightly wrong step. A wrongly large distance takes the policy off the data manifold in one update, and the further off it goes the better it looks to the critic that sent it there.

Modern generative policies sharpen the problem, because they remove the usual escape route. A policy that maps noise to a multimodal action through a learned sampler has no tractable likelihood, so the ratio-based toolbox — PPO, and anything else built on importance weights — is unavailable. This is true of diffusion and flow-matching policies and most acute for one-step samplers such as a drifting model (Deng et al., 2026), where a single network evaluation stands between noise and action. It can be recovered only by injecting noise into the deployed policy (Zhang et al., 2026), which changes what ships, or by assuming a denoising objective the policy does not have (McAllister et al., 2025) (Section 3.2). With likelihoods off the table, a critic is what remains — which makes how we consume it the whole question.

Both standard answers delegate the distance. Backpropagating $-Q$ through the actor, as the state-of-the-art residual fine-tuner DICE-RL (Sun & Song, 2026) does, makes every parameter step proportional to $\|\nabla_a Q\|$. The action-gradient family (Li et al., 2024; Psenka et al., 2025) never differentiates through the policy, but moves each sampled action to wherever a gradient step lands and regresses onto that — the same delegation, one level up. A controlled toy (Section 3.3) shows how bad this can get: with high-dimensional actions and scarce critic data, the backpropagated update drives true reward to zero while its own Q -estimate rises by four orders of magnitude. A behavior-cloning penalty does not save it, because a penalty in the loss must out-shout an unbounded gradient, and loses. Yet the gradient is not the problem. In the opposite regime, where the critic generalizes, it is the best signal available, and methods that discard it waste it. What is needed is a rule that keeps the direction in both regimes while capping how far any single step may carry the policy.

The lesson we draw is narrow and we state it as such: a critic may be trusted with the direction of a policy update, and with its length only up to a bound. We are not claiming the critic should be stripped of the step size entirely — in our own method it still sets it for the majority of sampled actions (Section 5.8), and that is fine. The claim is that the tail must be truncated, because that is where the critic is least constrained and where a single step can end the run. Discarding the direction, as critic-as-judge methods do (Peng et al., 2019; Wang et al., 2020), throws away the best signal available when the critic is trustworthy; letting the critic set the length hands it a quantity it was never trained to supply. Note that this indicts more than backpropagation. The action-gradient family — which improves actions along $\nabla_a Q$ and regresses the policy onto the improved actions (Li et al., 2024; Psenka et al., 2025) — keeps the direction but still leaves the distance to the critic, and is reported to degrade exactly as one would predict, drifting off the behavior distribution when the critic’s gradient is biased (Choi et al., 2026). What has been missing across both is not a better signal but a bound, and one placed where the harm occurs: in action space, not parameter space. Gradient clipping constrains the parameter step, which says nothing about how far the resulting action moves.

CAST supplies that bound. A frozen base proposes action particles through a residual head, and reward enters as a velocity field V_Q — by default the normalized critic gradient. Two guards then act before anything is regressed. First, each particle’s total displacement is clipped pointwise in action space, bounding how far the regression target may move — though not, as we are careful to establish in Section 4.1, how far the network’s output actually moves once those targets are amortized through shared parameters. Second, a restoring anchor field with a dead zone pulls the residual back toward the base whenever it strays beyond a radius ρ , leaving it untouched inside. Only then is the residual regressed onto the displaced targets, and because the update rewrites nothing in the replay buffer, stored transitions remain consistent with the dynamics that produced them. No likelihood is ever needed, and in the small-step limit the update recovers the backpropagated direction exactly (Proposition 1) — so the safety costs nothing asymptotically.

The construction assumes very little about what it is attached to. It needs a frozen map from noise to actions, K samples from it, and a critic; it never inspects the generator’s internals, never needs a likelihood, and never assumes a denoising objective. Consequently it applies unchanged to one-step, few-step, and many-step samplers, and we verify it on two of them. One economy is worth a remark but no more: a drifting model is itself pretrained by field-target regression (Deng et al., 2026), so on that base the fine-tuner happens to reuse

machinery the policy already has. That is convenience, not requirement, and no claim in this paper rests on it.

Because reward enters only as a field, the critic can also be consumed at lower resolution when its gradients cannot be trusted: transport toward critic-ranked top- k particles, or toward the cloud’s own $\exp(Q/\tau)$ -tilted density. Our experiments treat this as an explicit dial and map its regimes. On robomimic, where the critic generalizes, the gradient field is the resolution to use: from a drifting base at 0.877, CAST reaches 0.99 on can, and lifts square from 0.382 to 0.91–0.93 across seeds — ahead of the backpropagated control at the matched budget (0.87–0.91) and at parity with the flow-matching + DICE-RL state-of-the-art pipeline on both tasks, while the value-only sources trail far behind (~ 0.55 on square). On the eight hardest LIBERO-90 tasks, fine-tuned per task from one shared base with one shared recipe, CAST gains +3.9 to +23.9 points (mean +9.4) where the backpropagated actor consuming the same critics gains +0.9 and loses ground on the two highest-headroom tasks. The ordering is unchanged when the base is swapped for the stronger flow-matching policy, and CAST matches a critic-free policy-gradient baseline at equal budget without needing the noise injection that baseline requires. With a single critic shared across all eight tasks, no critic-based update rule — backpropagated or field-based, ours or the state of the art’s — improves on its own base, locating the binding constraint in per-task critic quality exactly as the toy predicts.

Our contributions:

- An action-space trust region for action-gradient policy improvement (Section 4). Improving actions along $\nabla_a Q$ and regressing the policy onto them is an established family (Li et al., 2024; Psenka et al., 2025; Choi et al., 2026), and we adopt that form unchanged. What the family lacks is anything holding the policy near the data, and the obvious candidate — a behaviour-cloning penalty applied at every step — is the wrong shape, because it opposes the critic in the region where the critic is reliable; measured, it lands at base level (0.549 and 0.601 against bases of 0.573 and 0.610). We therefore place the constraint where the harm occurs, in action space, and we test two forms of it: a pointwise clip on each particle’s displacement, and a restoring anchor that is silent within a radius ρ of the frozen base and engages only beyond it. We are deliberate about what this buys. Removing both is catastrophic — success falls to exactly 0.000 on both tasks — while either one alone recovers essentially all of the gain, and deleting the anchor entirely is not merely survivable but among our best settings (0.803 on t65). We therefore claim that some action-space constraint is necessary, and explicitly not that ours is the right one (Section 5.4). Two further choices separate our instantiation from the family: targets are formed from freshly drawn particles off a frozen base rather than from buffer actions mutated in place, so stored transitions stay consistent with the dynamics that produced them, and the update needs no likelihood.
- A controlled diagnosis of critic exploitation (Section 3.3): a two-regime toy showing the failure is real, regime-dependent, and not fixed by BC penalties — motivating trust regions in action space rather than loss space.
- Evidence that the form, not the signal or the step size, is what wins (Section 5): holding base, critic, data and budget fixed and varying only the actor update, CAST beats the backpropagated actor on both benchmarks and on both bases, lifts a weaker one-step base to match the stronger flow-matching + DICE-RL pipeline, and reaches parity with a critic-free policy-gradient baseline without modifying the deployed policy. Two controls close the obvious alternative explanations. Sweeping each method’s own step size an order of magnitude leaves the two in non-overlapping bands — the backpropagated actor never rises above its base at any learning rate, ours never falls near it at any reward step — so the gap is not tuning. And deleting only the critic’s gradient direction, keeping everything else, collapses performance below the untuned base on two separate tasks — so the gain is not simply caution.

2 Related Work

RL fine-tuning of generative robot policies. Policy-gradient fine-tuning of vision-language-action models with sparse task reward is now standard for autoregressive token policies (Li et al., 2025), but transfers poorly to continuous generative heads: likelihood ratios are intractable for diffusion and flow policies, and workarounds either reformulate the denoising chain as an MDP (Ren et al., 2025; Black et al., 2024) — a construction with no analog for one-step generators — or avoid likelihoods entirely (Park et al., 2025; Zhang et al., 2026; Mark et al., 2024). FPO (McAllister et al., 2025) is the strongest form of the likelihood-free ratio: it replaces $\pi_\theta/\pi_{\theta_{\text{old}}}$ with the exponentiated difference of per-sample conditional flow-matching losses, which the CFM-ELBO relationship makes a valid proxy, and it does so without perturbing the deployed policy. It is therefore the right policy-gradient method for a flow or diffusion policy — but the surrogate is defined by a denoising loss over a flow time τ , and a one-step drifting model has neither: its training objective is field-target regression, with no ELBO to bound. FPO also fine-tunes (or trains from scratch) the full policy on continuous-control suites, where we update a residual on a frozen base for manipulation. The two are complementary — FPO makes the ratio tractable, we make the critic safe to follow — and we regard evaluating FPO on a flow-matching base against our update as the clearest missing comparison in this paper. Closest to our setting, DICE-RL (Sun & Song, 2026) fine-tunes a frozen one-step distillation of a flow-matching policy by training a residual head with a backpropagated critic and a filtered BC anchor; it is our primary baseline and the control for our actor-update ablation, since we keep its critic, residual parameterization, and data pipeline unchanged.

Where this update sits. RL for diffusion and flow policies divides into four families (Choi et al., 2026): backpropagating through the denoising chain (Diffusion-QL, and DICE-RL’s residual actor); ratio-based methods that make a likelihood tractable (DPPO, FPO); Q -weighted regression; and action-gradient methods, which improve actions with $\nabla_a Q$ and then regress the policy onto the improved actions. CAST belongs to the last family, and we want to be exact about what is and is not new in it. The primitive of displacing actions along $\nabla_a Q$ and regressing onto the result is not ours: DIPO (Li et al., 2024) introduced it, updating replay-buffer actions by gradient ascent and fitting a diffusion policy to them, and QSM (Psenka et al., 2025) reaches the same signal differently, by matching the policy’s score to $\nabla_a Q$. What the family lacks — and what the survey notes none of its members supply — is any bound on how far that displacement may carry the policy, or any restoring force toward a behavior policy. That gap is also its documented weakness: action-gradient methods are reported to degrade when the critic’s gradient is biased, and to drift away from the behavior distribution.

Our contribution is precisely that missing piece, and our ablations measure it directly. The displacement is clipped pointwise in action space, a dead-zone anchor restores toward a frozen base, and the update is applied to a residual rather than by overwriting stored transitions (DIPO’s buffer rewrite makes the stored action inconsistent with the transition that produced it). Removing either guard reproduces the family’s failure inside our own system: performance falls below the untuned base (Table 9). Test-time guidance (Zhou et al., 2026) applies the same $\nabla_a Q$ signal without training at all, and is complementary.

Critic-as-compass versus critic-as-judge. That a learned Q -function’s action-gradient cannot be trusted is a recurring diagnosis: overestimation and narrow spurious peaks in TD3 (Fujimoto et al., 2018), extrapolation error in BCQ (Fujimoto et al., 2019), and spurious local optima that make sample-and-select beat gradient ascent (Jain et al., 2025). One response treats the critic only as a judge: weighted regression and sample-based improvement (AWR, CRR, MPO, IDQL, QT-Opt) (Peng et al., 2019; Wang et al., 2020; Abdolmaleki et al., 2018; Hansen-Estruch et al., 2023; Kalashnikov et al., 2018) tilt the policy toward $\pi \cdot \exp(Q/\tau)$ using only Q -values. Our method spans this spectrum rather than picking a side: the update form is judge-safe (bounded transport toward targets), while the reward field inside it can still be the analytic gradient where it is trustworthy — and our ablations map exactly when each choice wins.

Particle transport as a policy update. Regressing a sampler onto displaced particles is the amortized-SVGD idea of Soft Q-Learning (Haarnoja et al., 2017), with kernel repulsion preserving diversity as in SVPG (Liu et al., 2017); drifting models (Deng et al., 2026) make the same operator the pretraining objective of a one-step generator. Drifting Field Policy (Koo et al., 2026) is the closest prior method: it also performs RL on a drifting backbone by drifting fresh policy samples toward critic-ranked top- K candidates. CAST differs in three ways: we update a residual on a frozen base rather than the full policy, we transport with an explicit pointwise trust region and a restoring anchor field (the stability contract our toy shows is load-bearing), and our reward field is a dial — including the analytic $\nabla_a Q$ source that dominates when the critic generalizes — rather than a fixed ranking surrogate.

Residual RL and staying on-manifold. Learning a residual on a frozen base is a long-standing recipe for sample-efficient robot RL; DAWN (Ma et al., 2026) identifies the value cold-start and scale pathologies of residual critics that we also observe. The off-manifold drift our toy exhibits is the mechanism behind over-optimization and mode collapse in RL-finetuned generators (Black et al., 2024; Fan et al., 2023; Nakamoto et al., 2023), and forgetting in RL-finetuned language models is likewise governed by divergence from the base (Shenfeld et al., 2025) — which cannot even be measured without likelihoods, reinforcing update rules whose step size is bounded in action space by construction.

In sum, prior work either trusts the critic’s gradient and pays for it off-manifold, or discards it and pays on hard tasks; CAST is, to our knowledge, the first fine-tuner for one-step generative policies whose update form is safe under an arbitrary critic while still admitting the full first-order signal when it is available.

3 Problem Statement

3.1 Setup

We consider an episodic decision process with states s , action chunks $a \in \mathbb{R}^{H \times d}$ (H control steps of a d -dimensional action, treated as a single vector of dimension $S = Hd$), and sparse task-success reward. A one-step generative policy maps a state and a noise sample to an action in a single network evaluation,

$$a = \pi_0(s, z), \quad z \sim \mathcal{N}(0, I), \quad (1)$$

and is pretrained on demonstrations with a drifting-model objective (Deng et al., 2026): at each step, a population of policy samples is transported by a kernel velocity field toward the data distribution, and the network is regressed onto the transported targets. We write $\mathcal{V}(x \mid p^+, p^-)$ for this transport field acting on query particles x , attracted to particles of p^+ and repelled from particles of p^- (Section 4.2 recalls its exact form).

Following residual fine-tuning practice (Ma et al., 2026; Sun & Song, 2026), we freeze the base and learn a residual head on top of it,

$$a = \pi_0(s, z) + r_\theta(s, z), \quad (2)$$

together with an ensemble critic $Q_\phi(s, z, a)$ trained by temporal difference with a conservative (CQL-style) penalty (Kumar et al., 2020). The critic training is standard and identical for every method compared in this paper; the object of study is solely the actor update — how the policy consumes the critic.

3.2 Two structural failure modes

No likelihoods. Policy-gradient fine-tuners (PPO/GRPO-style, e.g. SimpleVLA-RL (Li et al., 2025)) require $\log \pi(a \mid s)$. A one-step generator defines its action distribution implicitly through the noise map Equation (2); its likelihood is intractable, and unlike multi-step diffusion policies there is no chain-MDP reformulation to fall back on (Ren et al., 2025; Park et al., 2025). Two routes recover a ratio, and neither is available to a one-step drifting policy. The first modifies the policy — adding fixed-variance Gaussian noise to its output so that a surrogate log-density exists (Zhang et al., 2026) — so the deployed policy is no

longer the deterministic one-step map that was pretrained; we include this surrogate as a baseline and it is a competitive one. The second leaves the policy untouched but requires it to be a flow model: FPO (McAllister et al., 2025) forms a PPO-clip ratio from differences of per-sample conditional flow-matching losses, exploiting the CFM-ELBO relationship. That construction needs a denoising loss defined over a flow time τ ; a drifting model has no τ and is trained by field-target regression, which carries no such ELBO link, so the surrogate does not exist for it. On a flow-matching base FPO is applicable and would be the natural policy-gradient comparison — a comparison we do not run here. The update we develop needs no likelihood, surrogate or exact, on either base.

Letting the critic set the distance. The remaining updates all use the critic’s action-gradient, and all let its magnitude decide how far the policy moves. Backpropagating $-Q$ through the actor, as in the DDPG lineage and in DICE-RL’s residual actor (Sun & Song, 2026), does so explicitly:

$$\nabla_{\theta}\mathbb{E}[-Q] = -\nabla_a Q \cdot \partial a / \partial \theta, \quad (3)$$

so a steeper critic produces a larger parameter step. The action-gradient family — which improves each sampled action by a gradient step and regresses the policy onto the improved action (Li et al., 2024; Psenka et al., 2025) — avoids differentiating through the policy, but inherits the same coupling one level up: the improved action sits wherever $a + \alpha \nabla_a Q$ lands, so the displacement is again proportional to the critic’s steepness. Reported failures of that family match the prediction, degrading under biased critic gradients and drifting off the behavior distribution (Choi et al., 2026).

Why is distance the risky thing to delegate? A critic is trained to rank actions near its data. Away from that data it is unconstrained, and can be steep and wrong at once (Fujimoto et al., 2018; 2019; Jain et al., 2025). The two quantities then fail asymmetrically. A slightly wrong direction costs a slightly wrong step. A wrongly large distance costs the data manifold: the policy is carried off it in one update, and the better it then scores under the critic, the worse it actually is.

The important word is persistent. The individual steps need not be large; what matters is that they keep pointing the same way. The critic’s extrapolated optimum does not move as the policy approaches it, so each update advances toward the same place and nothing returns the policy. This distinction turns out to decide which remedy works: limiting the size of a step does not stop a drift assembled from permitted steps, whereas a force that returns the policy does (Section 5.4).

3.3 A diagnostic experiment

To see the failure cleanly we build the smallest setting that contains it: a bandit whose expert data has two modes, a one-step policy pretrained on that data with the drift operator, and a critic trained only on actions the expert actually takes. The critic is therefore accurate on the data and unconstrained everywhere else — the situation every offline-trained critic is in. We then fine-tune the same base with two update rules and vary one thing: the action dimension, which controls how much of the space the critic has never seen.

When the critic is reliable, follow it. With 2-D actions and dense critic data, backpropagation is the best update (0.79 true reward, versus 0.53 for a value-only field update and 0.50 for the base). Nothing is wrong with using a critic’s gradient when the critic knows what it is talking about.

When it is not, backpropagation is catastrophic — and a BC penalty does not save it. With 16-D actions and 60 critic training points, the same update drives true reward to zero while the critic’s own estimate of that policy rises by four orders of magnitude: the policy has walked into a region the critic scores highly and has never seen. Adding the standard behavior-cloning penalty changes nothing (0.00 as well). This is the diagnostic result of the paper: the penalty is a term in a loss, and it is competing against a gradient of unbounded magnitude. It loses. Figure 1 shows the geometry — the policy is not wandering, it is being pulled, quickly and directly, toward the critic’s hallucinated optimum.

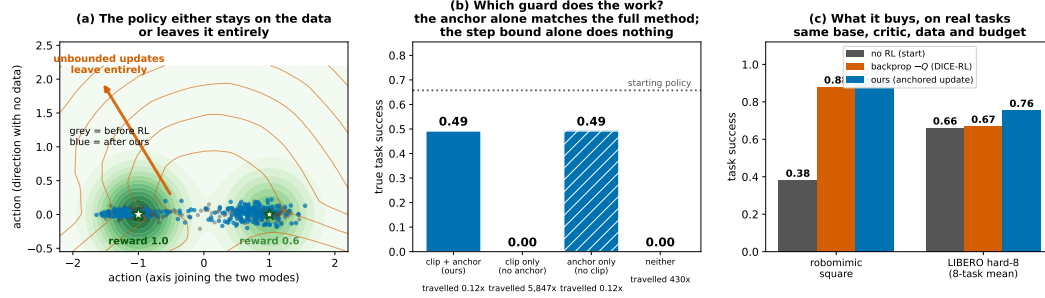


Figure 1: Which guard keeps a critic-driven update on the data? Panels (a)–(b) use a diagnostic bandit: 16-D actions, two demonstration modes of reward 1.0 and 0.6, and a critic deliberately fit to only 60 expert actions so that it is wrong away from them. We use it because the entire action space can be drawn, not because it resembles a robot task. (a) A slice of that space. Green is the true reward, thin orange lines are what the critic believes, and the two part company away from the data. Our update (blue) leaves the policy’s actions on the demonstrated modes; unbounded updates exit the frame entirely. (b) The two guards, switched on and off independently. A step bound on its own does nothing — 0.00 success, and $5,847\times$ the data’s spread away, no better than using no guard at all — because a drift assembled from many permitted small steps is still a drift. The dead-zone restoring force on its own reproduces the full method exactly ($0.49, 0.12\times$). This identifies the mechanism: what matters is the pull back toward the starting policy, not the limit on step size. (c) What that buys on real benchmarks, with base, critic, data and budget held fixed: robomimic square rises from 0.38 to 0.92, and the eight hardest LIBERO-90 tasks gain +9.4 points against +0.9 for the backpropagated actor. Note that in (b) no configuration exceeds the untuned base (0.66): the diagnostic is built to separate the guards, not to show improvement, which is the job of (c).

Which fix actually works. The natural cure is to limit how far the policy may move per update, and we tried it. It does not work: with the step bound alone the policy still ends at zero true reward, $5,847\times$ the data’s own spread away — no better than applying no guard at all. The reason is visible in Figure 1(a). The critic’s imagined optimum does not recede as the policy approaches it, so every permitted step points the same way, and a drift assembled from small permitted steps arrives just as surely as one large one. A speed limit is not a boundary.

What works is a restoring force that is silent within a radius ρ of the base and engages beyond it. On its own, with no step bound at all, it reproduces the full method exactly (0.49 true reward, $0.12\times$ the data spread), and adding the step bound on top changes nothing (Figure 1b).

Three conclusions carry into the method. First, the critic’s gradient is the best signal available where it generalizes and a liability where it does not, so a useful update keeps the direction. Second, what protects the policy is a force that returns it, not one that slows it: containment is a statement about where the policy ends up, not how fast it travels. Third — and this is why the dead zone rather than a plain penalty — the restoring force must be absent near the base, or it spends its strength opposing the critic in the one region where the critic is reliable. A constant behaviour-cloning penalty does exactly that, and in the same setting it also ends at 0.00.

4 Method

Section 3.3 left us with a concrete requirement: follow the critic where it has data, and prevent the policy from wandering to where it does not. Crucially, this is a requirement about where the policy ends up, not about how fast it travels — a distinction the experiments turn out to hinge on.

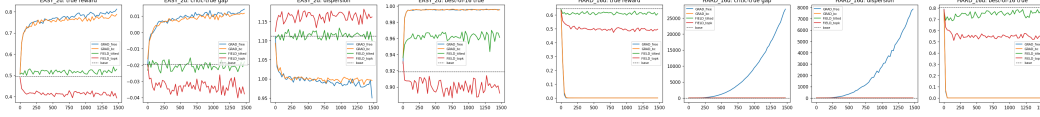


Figure 2: The two-regime toy (3 seeds). Left, easy regime (2-D actions, dense critic data): the critic’s gradient generalizes and backpropagation ($-Q$, with or without a BC anchor) is the best update. Right, hard regime (16-D actions, 60 critic points): the same updates collapse to zero true reward while their internal Q -estimate explodes — the BC anchor does not help — whereas the bounded field-target update stays on-manifold and holds the base policy. A useful fine-tuner must keep the gradient where it is trustworthy and survive where it is not.

The vehicle for that bound is an existing primitive, which we adopt rather than claim: move action samples along a velocity field, then regress the network onto the moved samples. This is how the action-gradient family improves a policy (Li et al., 2024; Psenka et al., 2025). What that family does not do, and what this section contributes, is constrain the displacement: in DIPO the improved action is wherever the gradient step lands, and the policy is then regressed onto it.

Nothing below is specific to a policy class. The update needs a frozen map from noise to actions, the ability to draw K samples from it, and a critic; it never needs a likelihood, a denoising objective, or access to the generator’s internals. It therefore applies to a one-step map, a few-step flow-matching sampler, or a many-step diffusion policy without modification, and we verify two of these empirically (Section 5.7). One convenience is worth noting without leaning on it: a drifting model is itself pretrained by field-target regression (Deng et al., 2026), so on that particular base the fine-tuner reuses machinery the policy already has. That is an implementation economy, not a requirement, and none of our claims depend on it.

The contribution is what happens between the critic and the regression: the proposed displacement is constrained in action space, on the target, before anything is fitted. The critic proposes an improvement; an explicit action-space trust region limits how much of that proposal the actor is asked to learn; only the constrained target is regressed. This is the difference from the backpropagated control, which admits the same proposal into a loss and lets it compete there against a regularizer (Section 4.4).

We instantiate the constraint two ways and claim neither over the other: a pointwise clip on each particle’s displacement, and an anchor field V_{anc} — a pull back toward the frozen base that is silent while the residual stays within a radius ρ and engages only outside it (Section 4.3 gives its form). The anchor is not an ad hoc construction: it is the negative gradient of a hinged radial potential, and at $\rho = 0$ it reduces exactly to the quadratic residual regularizer the backpropagated control already uses. The dead zone is thus a hinge generalization of standard L_2 residual regularization — which makes ρ an explicit interpolation between the loss-space regularizer of prior work and the constraint we advocate, and lets us measure the two against each other inside one implementation (Section 5.5).

The reward field V_Q says which direction improves each action, and its magnitude enters the target before either bound acts. We do not attribute the method’s behaviour to any single one of these components: what the experiments establish is that some action-space bound must be present (without one, success is exactly 0.000), and that within a wide band its particular form is not distinguishable at this budget (Section 5.4).

Neither field is a loss term; both are displacements applied to actions before the regression. This does not make the anchor invincible — where $\|V_Q\| > \|V_{\text{anc}}\|$ the net displacement still points outward — but it means the two compete as bounded magnitudes in action space rather than through an unbounded parameter gradient, as a behaviour-cloning penalty must (Section 3.3). No likelihood is required, and no critic gradient is ever backpropagated into the actor.

[Figure reserved: method overview.] Left: the frozen base π_0 maps noise particles z_i to a cloud of actions; the residual r_θ shifts them. Middle: the three velocity fields at each particle — clipped critic gradient V_Q (with the top- k / tilted alternatives inset), dead-zone anchor V_{anc} — and the total-motion clip δ_{tot} . Right: the regression of r_θ onto the displaced targets. To be drawn (TikZ or Figma).

Figure 3: CAST overview: the critic proposes a displacement, an action-space bound limits it, and the residual head is regressed onto the bounded targets. The base is any frozen noise-to-action map. **[TODO: draw]**

4.1 The field-target update

One update works as follows. For each state in the batch we draw K noise samples and form the current action particles $a_i = \pi_0(s, z_i) + r_\theta(s, z_i)$, treated as constants (detached). Each particle is displaced by a combined velocity,

$$\tilde{r}_i = \text{sg}\left[r_i + \text{clip}_{\delta_Q}(V_Q(s, z_i, a_i)) + V_{\text{anc}}(r_i)\right], \quad \mathcal{L}_{\text{actor}} = \frac{1}{K} \sum_i \|r_\theta(s, z_i) - \tilde{r}_i\|^2, \quad (4)$$

where $\text{sg}[\cdot]$ is stop-gradient and $\text{clip}_\delta(v) = v \cdot \min(1, \delta/\|v\|)$ rescales a field vector to norm at most δ .

In words, one update does four things. (1) Propose: draw K noise samples and push them through the frozen base plus the current residual to get a cloud of candidate actions for this state. (2) Score and displace: ask the critic which direction improves each candidate, clip that direction to length δ_Q , and add the anchor’s pull if the residual has strayed past its radius; clip the resulting total displacement to δ_{tot} . (3) Freeze the targets: the displaced actions are detached — they are data now, not a computation graph. (4) Regress: train the residual head to output those displaced targets. The critic is trained separately (Section 4.4); it is never differentiated through the actor, only queried.

Why this primitive, for this policy class. Field-target regression is not merely a safe update — it is the operation the network was shaped by. A step-distilled generator fine-tuned under a mismatched objective drifts from the distribution its one-step map encodes (Jiang et al., 2026; Yang et al., 2024); by keeping fine-tuning inside the pretraining operator family, CAST changes only where the targets sit, not what kind of function the network is asked to represent.

What the clip does and does not guarantee. The clip bounds target transport, and only that:

$$\|\tilde{r}_i - r_i\| \leq \delta_{\text{tot}} \quad \text{for every particle } i. \quad (5)$$

It is tempting to read this as a bound on how far the policy moves, and an earlier version of this work did. That inference is invalid. The targets are amortized through a shared network by a gradient step, so to first order the realized change at particle i is

$$\Delta r_i \approx \alpha \sum_j J_i J_j^\top (\tilde{r}_j - r_j), \quad J_i = \partial r_\theta(s, z_i) / \partial \theta, \quad (6)$$

and the Jacobian coupling across particles carries no guarantee that $\|\Delta r_i\| \leq \delta_{\text{tot}}$. This is precisely the objection we raise against parameter-space gradient clipping (Section 3.2), and it applies to target-space clipping too: bounding what the network is asked for does not bound what it does. A per-particle guarantee would require an output-space proximal step, exact interpolation, or a tabular actor — none of which we impose.

The distinction matters, but it does not undermine the construction, because a per-particle guarantee is not what the update needs. A drift assembled from many individually permitted steps is still a drift, since the critic keeps pointing the same way and nothing returns the policy — which is why the constraint that matters is one whose strength grows with the displacement already accumulated, rather than one that merely rate-limits each step. What we establish empirically is narrower and sufficient: with no action-space bound of any kind the update destroys the policy, and with either bound present it does not (Section 5.4).

4.2 The reward field V_Q

The default (and headline) reward field is the normalized critic gradient evaluated at each particle,

$$V_Q^{\text{grad}}(s, z, a) = \eta_Q \frac{\nabla_a \bar{Q}_\phi(s, z, a)}{q_{\text{scale}}}, \quad q_{\text{scale}} = \mathbb{E}[|\bar{Q}_\phi|], \quad (7)$$

where $\bar{Q}_\phi$ is the conservative ensemble aggregate. This is deliberately the same scalar the DICE-RL actor optimizes: in the small-step, unclipped limit, minimizing Equation (4) recovers the backpropagated update direction exactly (Proposition 1), so any performance difference is attributable to the update form — the anchored target regression — and not to a different reward signal.

The kernel transport operator. The lower-resolution reward fields below, the pretraining objective, and the distributional anchor variant all use the same drifting-model transport operator (Deng et al., 2026). Given query particles x , attractor particles p^+ (optionally weighted), and repulsor particles p^- , distances are normalized by a global scale and, for each kernel radius R in a small set, a soft mutual-matching affinity $A = \sqrt{\text{softmax}_{\text{row}}(-d/R) \odot \text{softmax}_{\text{col}}(-d/R)}$ converts matched attractors into pulls and matched repulsors into pushes; forces are summed over radii to give $\mathcal{V}(x | p^+, p^-)$ (full form in Appendix C). Intuitively, $\mathcal{V}$ moves a particle cloud so its density flows toward p^+ and away from p^- — it needs only samples of the target, never a density or a gradient.

Because V_Q enters Equation (4) only as a velocity, the critic can be consumed at lower resolution when its gradients are not trustworthy, using only Q -values:

- zeroth-order (top- k): rank the K particles by Q and set $V_Q = \mathcal{V}(a | \text{top-}k \text{ particles, all particles})$ — the pretraining kernel transport toward the critic’s preferred candidates;
- tilted: let every particle attract with weight $K \cdot \text{softmax}_i(\text{adv}_i/\tau)$, transporting the cloud toward its own $\exp(Q/\tau)$ -tilted density, the canonical improvement target of Peng et al. (2019); Abdolmaleki et al. (2018).

These are not competing methods but settings of a dial — how much of the critic’s local structure the update trusts: full first-order information, ranking only, or soft value weights. Section 5 maps the regimes: the gradient source dominates wherever the critic generalizes, while the value-only sources close the gap precisely in the weak-critic regime predicted by Section 3.3.

4.3 The anchor field

The kernel reward fields vanish beyond their largest radius and the gradient field carries no notion of the data manifold, so Equation (4) alone would let the residual random-walk. The anchor is a restoring field with a dead zone,

$$V_{\text{anc}}(r) = -\eta_{\text{anc}} \left(1 - \frac{\rho}{\|r\|} \right)_+ r, \quad (8)$$

which is inert inside the trust radius ρ and pulls back only the excess residual norm beyond it. Equivalently, Equation (8) is the negative gradient of a hinged radial potential,

$$U(r) = \frac{1}{2} \eta_{\text{anc}} (\|r\| - \rho)_+^2, \quad V_{\text{anc}}(r) = -\nabla_r U(r), \quad (9)$$

Algorithm 1 CAST: one update (V_Q^{grad} variant)

Require: frozen base π_0 ; residual r_θ ; critic Q_ϕ ; steps $\eta_Q, \eta_{\text{anc}}$; clips $\delta_Q, \delta_{\text{tot}}$; dead zone ρ ; particles K

- 1: sample batch of states s ; draw $z_i \sim \mathcal{N}(0, I)$, $a_i \leftarrow \pi_0(s, z_i) + r_\theta(s, z_i)$ $\triangleright$ no grad
- 2: $V_i \leftarrow \text{clip}_{\delta_Q}(\eta_Q \nabla_a \bar{Q}_\phi(s, z_i, a_i) / q_{\text{scale}})$ $\triangleright$ or top- k / tilted kernel field
- 3: $V_i \leftarrow V_i - \eta_{\text{anc}}(1 - \rho / \|r_i\|)_+ r_i$ $\triangleright$ dead-zone anchor, Equation (8)
- 4: $\tilde{r}_i \leftarrow \text{sg}[r_i + \text{clip}_{\delta_{\text{tot}}}(V_i)]$ $\triangleright$ policy moves $\leq \delta_{\text{tot}}$, always
- 5: $\theta \leftarrow \theta - \alpha \nabla_\theta \frac{1}{K} \sum_i \|r_\theta(s, z_i) - \tilde{r}_i\|^2$
- 6: update ϕ : ensemble TD + CQL penalty $\triangleright$ unchanged from Sun & Song (2026)

so it removes exactly the radial excess and nothing else. Setting $\rho = 0$ gives $U(r) = \frac{1}{2} \eta_{\text{anc}} \|r\|^2$ — the quadratic residual regularizer of the backpropagated control — so the dead zone is a hinge generalization of that penalty, with ρ interpolating between them. Unlike a $\lambda \|r\|^2$ penalty — which competes with an unbounded gradient and loses (Section 3.3) — Equation (8) composes with the clipped reward field as a velocity, so the residual scale is set geometrically rather than by a loss-weight arms race. Heuristically, if the reward field were stationary, radially outward and saturated at $\|V_Q\| = \delta_Q$, balancing it against the anchor outside the dead zone gives $\delta_Q = \eta_{\text{anc}}(\|r\| - \rho)$, i.e. $\|r\| = \rho + \delta_Q / \eta_{\text{anc}}$. We use this only as a scale estimate when choosing η_{anc} : a learned critic’s field varies in direction and magnitude with r , so there may be several fixed points, cycles, or none, and we claim no unique equilibrium. pointwise pull $-\eta_{\text{anc}} r$, whose fixed point reproduces the DICE-RL anchor; the dead zone matters exactly when the reward field is weak (Section 5, LIBERO).

4.4 Relation to the backpropagated actor; critic

Proposition 1 (Small-step equivalence). At the current parameters, the gradient of Equation (4) is $\nabla_\theta \mathcal{L}_{\text{actor}} \propto -(\text{clip}_{\delta_Q}(V_Q) + V_{\text{anc}})^\top \partial r_\theta / \partial \theta$. With $V_Q = V_Q^{\text{grad}}$, no clipping ($\delta_Q, \delta_{\text{tot}} \rightarrow \infty$) and $\rho = 0$, this is the DICE-RL residual actor gradient with anchor weight η_{anc} .

CAST is therefore a strict generalization of its control: the new degrees of freedom are the pointwise action-space clip, the dead-zone anchor, and the resolution dial on V_Q — each of which we ablate.

Remark (not just clipped gradient ascent). One might object that Equation (4) with V_Q^{grad} is “DDPG with clipping.” The distinction is where the bound lives. Clipping ∇_θ bounds a parameter step, whose effect on the policy output is unknown and state-dependent; Equation (4) bounds the policy’s motion at every particle, which is the quantity that off-manifold failure depends on (Section 3.3). Moreover, only the regression form makes the two other components expressible at all: a dead-zone anchor and a value-only V_Q have no loss-gradient counterpart. The ablations in Section 5.5 test each distinction in isolation. The critic is unchanged from DICE-RL: ensemble TD with a CQL-style penalty pushing Q down at policy actions and up at data actions; field construction never leaks gradients into the critic (gradients w.r.t. actions only). Algorithm 1 summarizes one joint update.

5 Experiments

We test four claims. The first asks how the method compares to what a practitioner could use today. The second is the causal claim, and is the paper’s main result. The third and fourth ask which part of the method is responsible for it.

H1 CAST matches the strongest existing fine-tuners. The methods that, like it, let the critic displace actions — but with no action-space constraint — do not improve their base at all.

H2 Change only the actor update, and CAST gets more out of the same critic than backpropagating $-Q$ does.

H3 Take the action-space constraint away, and the update destroys the policy.

H4 The constraint has to act on the action target. The same shape placed in the loss, or on the parameter gradient, does not work.

Two supporting analyses follow: how much of the critic the update should consume (Section 5.6), and whether the result depends on the generator class the update is attached to (Section 5.7).

Throughout, “—” means not measured. No cell is inferred from a related run.

5.1 Setup

Benchmarks. Two, chosen because their critics differ. robomimic can and square are state-based, single-task, and densely covered, so the critic generalizes well; we report 300-episode evaluations averaged over the last three checkpoints, on 3 seeds. The eight hardest tasks of LIBERO-90 are image-based and sparsely rewarded, so the critic is far weaker; we report powered evaluation — 100 rollouts $\times$ 3 evaluation seeds. Both run inside the official DICE-RL harness with only the actor update replaced, so critic, replay, demonstrations and evaluation are byte-identical to the baseline. [TODO: appendix table: env details, horizons, demo counts, budgets.]

How we report success. Fine-tuners in this literature start from different pretrained policies, so absolute success confounds the fine-tuner with its base. Where methods do not share a base we give each method’s own base next to its result, and compare $\Delta = \text{fine-tuned} - \text{its own base}$. Only in Section 5.3, where every row is literally the same checkpoint, do we compare absolute numbers.

Methods. Every method on the drifting base shares the residual parameterization, critic, data and budget, so only the actor update varies.

- CAST — Equation (4) with the gradient reward field V_Q^{grad} .
- Backprop actor — the DICE-RL residual actor (Sun & Song, 2026): backpropagated $-Q$ with a filtered behaviour-cloning penalty, run on our drifting base. It reads the identical critic through the identical residual head, so any gap is attributable to the update form. Note that it already carries two constraints of its own: the BC penalty in its loss, and gradient-norm clipping at 1.0 on the actor. Both matter in Section 5.5.
- FM + DICE-RL — the published pipeline, unmodified, on its own flow-matching base.
- Classic diffusion-RL fine-tuners (DIPO, QSM, DQL, IDQL, AWR) — official implementations, unmodified, from their own diffusion base at their own published budget.
- CAST (top- k) and (tilted) — the same update with the reward field built from the critic’s rankings or its soft values instead of its gradient (Section 5.6).

Hyperparameters are shared across CAST and its variants, one recipe per benchmark and never per task: $\delta_{\text{tot}}=0.15$, $\eta_{\text{anc}}=1.0$, $\tau=1$, $k=4$, with reward step $\eta_Q=0.5$ on robomimic and $\eta_Q=1.0$ on LIBERO, and a dead-zone radius $\rho=0.05$ in dimension-independent rms units on LIBERO.

5.2 H1: where the method stands among existing fine-tuners

Claim. CAST matches the strongest existing fine-tuners, and the methods closest to it — those with no action-space constraint — do not improve their base at all.

Verdict. Both halves hold. We match the strongest pipeline rather than beating it, and we say so.

Table 1: Fine-tuners on robomimic square, each from its own pretrained base. The classic diffusion-RL methods run from official implementations without modification, at their published settings, which give them $300 \times 400 \times 50 \times 4 \approx 24\text{M}$ environment steps against our 192K — roughly 125 $\times$ our interaction. This is deliberately generous to them and is not a matched-budget comparison. They are also robomimic-only: their official implementations do not support LIBERO, so the LIBERO comparison in Table 3 is carried by FM + DICE-RL and GRPO instead. Success uses the same 300-episode protocol throughout. “—” = not measured; we have not run the flow-matching base on square, so its Δ is unavailable.

method	policy class	own base	fine-tuned	Δ
DIPO (Li et al., 2024)	diffusion	0.279	0.283	+0.004
QSM (Psenka et al., 2025)	diffusion	0.279	0.000	−0.279
DQL	diffusion	0.279	0.000	−0.279
IDQL	diffusion	0.279	0.041	−0.238
AWR	generic	0.279	0.407	+0.128
FM + DICE-RL (Sun & Song, 2026)	flow matching	—	0.923–0.934	—
CAST	one-step drifting	0.382	0.912–0.930	+0.53–0.55

Against the published flow-matching pipeline we reach parity, not superiority: 0.912–0.930 against 0.923–0.934 across three seeds. What we can say is that we reach it from a base of 0.382.

The two finished rows above the line are the ones that matter for this paper. DIPO ends at 0.283 against its own base of 0.279 — four tenths of a point after 24M environment steps, inside evaluation noise. QSM collapses to exactly 0.000 in all six evaluation blocks.

We checked our own setup before reporting a collapse. Every hyperparameter matches the published configuration ($n_{\text{critic warmup}}=5$, actor/critic learning rates $10^{-5}/10^{-3}$, q_{grad} coefficient 10, $\tau=0.005$, buffer 5000, replay ratio 16, batch 1000), and it gets more training than published, not less (300 against 201 iterations). The failure is immediate rather than gradual: base level (0.284) at iteration 0, zero by iteration 10, unchanged for the next 290. Both losses converge (actor 0.002, critic 0.01) rather than diverging. That is the signature of a degenerate fixed point — once the policy stops succeeding every reward is zero, the critic correctly learns $Q \approx 0$ everywhere, and $\nabla_a Q$ carries no signal with which to recover. One caveat: our diffusion base scores 0.279 where the original authors report roughly 0.4, and a weaker base implies a worse early critic, so we do not claim this reproduces any published QSM number.

The five rows split cleanly, and the split is the point. Group them by what the critic is allowed to do. DIPO, QSM and DQL let the critic displace actions — through an action-space gradient step, a score matched to $\nabla_a Q$, or backpropagation through the policy — with nothing bounding the displacement. All three fail outright: two at exactly 0.000, one at +0.004 after 24M environment steps. AWR lets the critic only weight actions the policy already produced, and it is the one method that improves (0.407, +0.128): with no displacement there is nothing for a missing bound to break. IDQL sits between — it never displaces either, but its Q-based action selection still exploits critic error, and it ends below its base (0.041). The failure is therefore not “critic-guided fine-tuning is unstable,” and not our harness: it tracks precisely whether the critic controls where actions move. That is the quantity our constraint bounds. The displacing methods are the unconstrained corner of our own design space, measured externally in their authors’ code at 125 $\times$ our interaction — and Section 5.4 reproduces exactly their behaviour inside our system by switching our constraint off. What separates CAST from that family is not the signal, the budget, or the policy class; it is the bound. AWR’s positive result sharpens rather than weakens the claim: consuming the critic safely was never the hard part — weighting schemes have always done so — but they leave the critic’s most informative output, its action gradient, on the table (Section 5.6 measures what that costs on the harder task).

Table 2: Matched actor-update comparison on robomimic (300-episode evaluations, last-3-checkpoint average, per seed). Every row shares the same base checkpoint, critic, data, replay and budget; only the actor update differs. [†] these two runs ended before 20K steps, so their last-three averages are taken at 17–19K (s42) and 14–16K (s43); both curves are flat there, so the earlier window if anything favours the baseline.

actor update	can			square		
	s42	s43	s44	s42	s43	s44
none (frozen base)		0.877			0.382	
backprop $-Q$	0.970 [†]	0.946 [†]	0.940	0.870	0.909	0.861
field-target + constraint	0.988	0.993	0.978	0.912	0.930	0.929

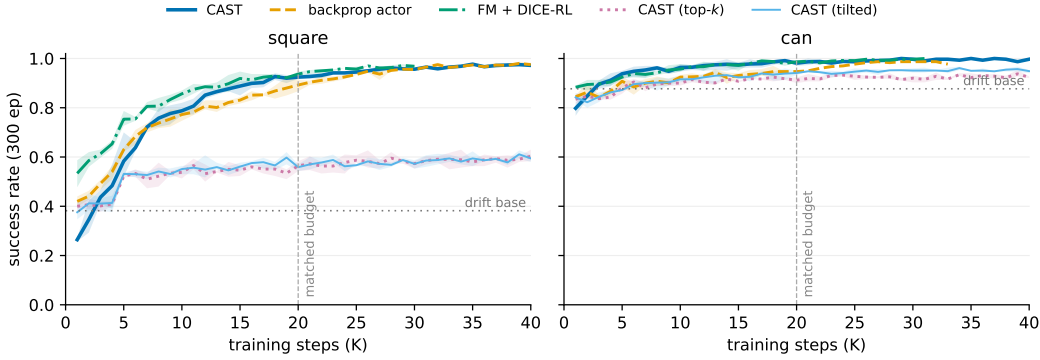


Figure 4: robomimic learning curves (300-episode evaluations every 1K steps; mean and min-max band over seeds; dashed line marks the matched 20K budget). On square CAST overtakes the backprop actor by 7K steps and holds a 4–7-point lead through the matched budget; the value-only variants flatline near 0.55. On can both approach the ceiling. [TODO: regenerate with the s44 and can-baseline reruns when they finish.]

5.3 H2: the update form is what converts critic signal into success

Claim. Change only the actor update, and CAST gets more out of the same critic than backpropagating $-Q$ does.

Verdict. Holds on both benchmarks, on every seed, on both bases. This is the paper’s main result.

Can is close to saturated and mainly checks that nothing breaks: CAST reaches 0.978–0.993 from a base of 0.877, the backprop actor 0.940–0.970. Square is the discriminative task. Its base solves only 0.382 of episodes; CAST adds 55 points to reach 0.91–0.93 on every seed, where the backprop actor reaches 0.86–0.91. Same critic, same data, same budget, different update: the field-target form extracts more, on every seed of both tasks.

Figure 4 shows the difference is in speed as much as in endpoint. CAST reaches 0.91 on every seed by 15K steps; the backprop actor needs another 5–10K steps to get there, trails at the matched budget on every seed, and catches up only near 40K where both converge to 0.97–0.99. A bounded update buys a faster, never-worse path to the same asymptote.

LIBERO hard-8. Here we deploy CAST as a routing system: one frozen multitask drifting base, one small per-task residual head and critic fine-tuned on each task’s own interactions, and task-ID routing at deployment. This is the architecture every published LIBERO RL fine-tuner effectively uses. One recipe is shared across all eight tasks with no per-task tuning.

The verdict spans all eight tasks. CAST gains +3.9 to +23.9 points per task (mean +9.4 at $\eta_Q=2$, +7.7 at $\eta_Q=1$) and improves on the base for every one. The backprop actor, reading

Table 3: LIBERO hard-8, per-task fine-tuning from the shared drifting base. Powered evaluation: 100 rollouts $\times$ 3 evaluation seeds; CAST rows average the last three training checkpoints, and two training seeds on t32/t53/t65. One recipe across all tasks. Across the five tasks trained with two seeds the spread is 0.000, 0.003, 0.004, 0.019 and 0.089 (t75, t73, t53, t32, t65) — negligible except on t65, where only gaps above ~ 9 points are meaningful. The lower block repeats the comparison on a flow-matching base (Section 5.7); its blanks are unrun cells, and a row mean is withheld until all eight are measured.

	t8	t21	t32	t53	t65	t73	t75	t81	mean
drift base (no RL)	0.818	0.690	0.610	0.785	0.573	0.648	0.578	0.588	0.661
backprop actor	0.820	0.703	0.587	0.800	0.550	0.663	0.583	0.653	0.670
CAST ($\eta_Q=1$)	0.831	0.737	0.694	0.842	0.826	0.710	0.657	0.607	0.738
CAST ($\eta_Q=2$)	0.857	0.752	0.729	0.853	0.812	0.752	0.648	0.635	0.755
same protocol, flow-matching base									
FM base (no RL)	0.897	0.700	0.690	0.917	0.787	0.683	0.620	0.733	0.753
FM + DICE-RL	0.873	—	0.682	0.915	0.809	0.690	0.592	0.708	—
FM + GRPO	—	—	0.700	0.897	0.883	—	0.603	—	—
FM + CAST	—	—	0.738	0.897	0.878	—	0.608	—	—

Table 4: LIBERO hard-8 mean success under powered evaluation (100 rollouts $\times$ 3 evaluation seeds per cell; one column per training seed; 100 RL iterations). Neither base is moved by any fine-tuner.

method	s10000	s10001	s10002
drift base (no RL)	0.671 ± 0.014		
backprop actor (same base)	0.657	0.655	0.668
CAST (top- k)	—	0.684	—
CAST (tilted)	0.677	—	0.672
CAST	0.655	—	0.659
flow-matching base (no RL)	0.738 ± 0.022		
FM + DICE-RL	0.756	0.735	0.737

the same per-task critics, averages +0.9 and drops below base on the two highest-headroom tasks. The robomimic ordering transfers intact to the harder benchmark.

Two implementation details were load-bearing and are ablated in Section 5.8. The dead zone must be measured in dimension-independent rms units: in vector-norm units the base residual already exceeds ρ at LIBERO’s $S=112$ chunk dimension, so the anchor is active from the first update and freezes the residual. And the weaker critic gradient here rewards a doubled reward step, which the constraint makes safe to take: $\eta_Q=2$ lifts the mean from 0.738 to 0.755 and helps on six of eight tasks, with no setting on any task falling below its base.

Comparison to a critic-free policy gradient. On the one flow-matching task with real headroom, GRPO and CAST extract the same amount (0.883 and 0.878 on t65, both about +9.5) while the backprop actor manages +2.2. On t32 CAST leads (0.738 against 0.700). On the tasks where nothing gains, all three sit together. So the separation on this benchmark is not CAST against the world; it is update form against backpropagation-through-the-critic. CAST matches what a policy-gradient method extracts without needing the noise-injection surrogate that makes likelihood ratios definable for a one-step generator in the first place. [TODO: per-task figure; remaining FM cells and matched-budget FM + GRPO.]

Where H2 stops: the shared-critic control. The alternative to routing is one policy and one critic trained on all eight tasks jointly. This flips the critic’s situation — our diagnostics show the shared critic’s action-gradient collapsing on most tasks, with per-task $\|\nabla_a Q\|$ spanning 0.003 to 0.55.

Table 5: Constraint factorial on the frozen LIBERO recipe, powered evaluation (last-3 checkpoints). “Clip off” disables both displacement bounds (δ_Q and δ_{tot}); “anchor off” disables both restoring terms. The ($\times$, $\times$) row is the unconstrained action-gradient update — the form DIPO and QSM run (Table 1). “—” = not measured.

clip	anchor	configuration	t65 (base 0.573)	t32 (base 0.610)
$\times$	$\times$	unconstrained transport	0.000	0.000
$\checkmark$	$\times$	step containment only	—	—
$\times$	$\checkmark$	base-restoring only	—	—
$\checkmark$	$\checkmark$	full CAST	0.781	0.684

[Figure reserved: containment dynamics.] Left — success against training iteration for the four rows of Table 5 plus the backprop control. Right — distance of the policy from its frozen base, $D_{\text{base}}(t) = \mathbb{E}_{s,z} \|r_\theta(s, z)\|_{\text{rms}}$, and the fraction of particles outside the trust radius $P_{\text{out}}(t)$, for the same arms. The prediction is that the unconstrained arm’s D_{base} grows without bound as its success falls to zero, while every constrained arm holds D_{base} stationary.

Figure 5: Containment, measured rather than assumed. [TODO: runs launched.]

Nothing wins here. On the drifting base the backprop actor (0.655–0.668), CAST (0.655–0.659) and every resolution variant (0.66–0.70) all sit within noise of the untuned base (0.671 ± 0.014). The flow-matching pipeline looks stronger in absolute terms (0.735–0.756), but its own base already scores 0.738 ± 0.022 , so it too extracts essentially nothing; its apparent lead is inherited from pretraining. Applying our final recipe — every fix and hyperparameter behind the +9.4 — changes nothing here (0.668 against base 0.671).

This bounds H2 rather than contradicting it. The update form converts critic signal into success; where there is no usable signal there is nothing to convert. It is also worth noting that every drift-base arm preserves the base within noise, including the backprop actor — consistent with a critic that is action-blind rather than action-exploitable. There is no steep hallucinated gradient here to ride. Finally, a methodological note: single-seed 20-episode endpoints ranged 0.63–0.78 across these same arms, and powered evaluation collapses that spread to the base value. [TODO: per-task breakdown figure once the remaining seed evaluations land.]

5.4 H3: the action-space constraint is what makes the form viable

Claim. Take the action-space constraint away, and the update destroys the policy.

Verdict. The decisive cell is measured and unambiguous: success falls to exactly 0.000. The other three cells are running.

CAST bounds the target’s motion two ways: a pointwise clip on each particle’s displacement, and the dead-zone anchor. We claim neither over the other. The hypothesis is that some action-space constraint is necessary, so the experiment is the full factorial.

With no constraint at all, success is exactly 0.000 on both tasks, from bases of 0.573 and 0.610. The unconstrained action-gradient update destroys the policy. This reproduces inside our own system — our critic, our data, our budget — the behaviour DIPO and QSM show in their authors’ code at $125\times$ the interaction. That is the paper’s central negative result, and it is why we claim the constraint rather than the transport.

Table 6: Dead-zone radius sweep, single seed, powered evaluation (last-3 checkpoints). ρ varies the shape of the restoring field, not its presence. The genuinely unconstrained configuration is Table 5’s first row.

dead-zone radius ρ	t65 (base 0.573)	t32 (base 0.610)
0 (always-on penalty)	0.549	0.601
0.02	0.863	0.734
0.05 (recipe)	0.781	0.684
0.10	0.812	0.694
0.20	0.771	0.671
∞ (dead zone inert; weak pull + clip remain)	0.803	0.671

Figure 5 is where the mechanism becomes visible instead of inferred. Endpoint success tells us that constrained arms work; the residual trajectory tells us whether the policy was held near its base throughout, or merely happened to end near it.

The constraint’s shape is not critical; its presence is. Table 6 sweeps the dead-zone radius over two orders of magnitude. Every interior setting lands between 0.771 and 0.863 on t65 and between 0.671 and 0.734 on t32 — a spread no larger than this benchmark’s seed-to-seed variation.

The two endpoints behave differently, and the difference is the point. At $\rho=0$ the anchor becomes an always-on behaviour-cloning penalty and performance falls to roughly base level on both tasks: an always-on pull erases the residual. At $\rho \rightarrow \infty$ the dead-zone term goes inert, but this is not an unconstrained run — the weak pointwise restoring component and the displacement clip both remain, so it is a re-shaped constraint rather than an absent one. The genuinely unconstrained configuration is the first row of Table 5.

So a constraint must exist, and it must not be always-on. Between those two failure modes the setting is not critical, and we claim no preferred radius: $\rho=0.05$ sits inside a wide flat region, not at an optimum.

5.5 H4: the constraint has to act on the action target

Claim. The constraint has to act on the action target. The same shape placed in the loss, or on the parameter gradient, does not work.

Verdict. Two rows support it. The decisive row — giving the backpropagated actor our exact dead-zone shape inside its loss — is not yet run. We name it below as the experiment that would overturn this claim.

The most economical objection to this paper is that CAST is DICE-RL with a differently shaped regularizer, and that putting the constraint on the action target rather than in the loss is presentation rather than mechanism. Table 7 is built to test exactly that.

Two rows are filled and they agree. The first is DICE-RL exactly as published, and it is worth being precise about what it already contains: a behaviour-cloning penalty in the loss and gradient-norm clipping at 1.0 on every actor step of every run in this paper. It constrains in both of the two places we say are wrong, and it still reaches only 0.550 and 0.587 against bases of 0.573 and 0.610.

The $\rho=0$ row makes the same point from inside our own method, and the correspondence is exact rather than analogous: by Proposition 1, CAST at $\rho=0$ with the clips removed is the DICE-RL residual actor’s update. Give the constraint the always-on quadratic shape that prior work uses and the policy returns to base level. This is the quantitative form of the toy’s finding that a penalty inside a loss must out-shout an unbounded gradient, and loses.

What remains is the sharpest version of the objection, and we state it as an open cell rather than a result. DICE + hinge gives the backpropagated actor our dead-zone geometry inside its loss. It therefore shares CAST’s first-order gradient and differs only in the update form — multi-particle detached target regression against a single backpropagated loss. If it matches

Table 7: Where the constraint acts, at matched geometry. All rows share base, critic, data and budget. The first row is DICE-RL as shipped, which already constrains in two places at once — a behaviour-cloning penalty in the loss and gradient-norm clipping at 1.0 on the actor — so it is the joint loss-space/parameter-space entry rather than two separate rows. “DICE + hinge” gives that same actor CAST’s anchor potential $(\|r\| - \rho)_+^2$ as a loss term: by Proposition 1 it shares CAST’s first-order gradient and differs only in the update form. “—” = not measured.

constraint	where it acts	t65 (base 0.573)	t32 (base 0.610)
L_2 penalty + grad clip (DICE-RL)	loss + parameters	0.550	0.587
hinge $(\ r\ - \rho)_+^2$ penalty	loss	—	—
always-on pull (CAST, $\rho=0$)	action target	0.549	0.601
hard projection $\Pi_{\ r\ \leq \rho}$	action target	—	—
dead-zone anchor (CAST)	action target	0.781	0.684

Table 8: Deleting the critic’s gradient direction. Square is the 3-seed range from Table 2’s protocol; LIBERO cells are powered evaluations on the frozen recipe.

reward field V_Q	square (3 seeds)	t65 (base 0.573)	t32 (base 0.610)
gradient (CAST)	0.912–0.930	0.781	0.684
top- k (ranking only)	0.525–0.562	0.577	0.341
tilted ($\exp(Q/\tau)$)	0.560–0.584	0.697	0.378

CAST, the geometry is the whole story and the form is presentation. If it fails where the base-restoring row of Table 5 succeeds, the form is the mechanism. The experiment is two-sided by construction and we will report it either way.

5.6 How much of the critic should the update consume?

Claim. The critic’s gradient direction matters. CAST is not winning just by moving less.

Verdict. Holds on both benchmarks and both tasks.

Because reward enters only as a field, the critic can be consumed at lower resolution: transport toward its top- k ranked particles, or toward the cloud’s own $\exp(Q/\tau)$ -tilted density. Both delete the gradient direction and change nothing else.

The answer is the same wherever we ask it. On robomimic, where the critic generalizes, the value-only fields preserve can (0.91–0.95) but collapse on square, despite verified-healthy residuals and longer training. On LIBERO the same two substitutions fall to 0.577 and 0.697 on t65, and to 0.341 and 0.378 on t32 — not merely below the full recipe but far below the untuned base, deficits of 34 and 31 points.

Two benchmarks, two bases, one conclusion: the ranking information a critic can supply is not a substitute for its gradient. CAST keeps the critic’s most informative signal and bounds only how far that signal may carry the policy in one step. Conservatism alone would not produce these numbers — it would produce the base.

5.7 Generality across generator classes

Claim. The update needs only a frozen noise-to-action map, so it transfers to other generator types.

Verdict. Holds on the four flow-matching tasks measured so far. Four remain, so we withhold a row mean.

On its own, stronger flow-matching base the published pipeline extracts essentially nothing per task: FM + DICE-RL scores -0.8 , -0.2 , $+2.2$ and -2.8 points relative to that base on t32, t53, t65 and t75 (mean -0.4) — the same near-zero extraction its actor shows on the

Table 9: Recipe sensitivity: leave-one-out on the frozen LIBERO recipe, powered evaluation (last-3 checkpoints). Each row changes exactly one component; Δ is relative to the full recipe. Resolution-dial rows are in Table 8. “—” = not measured.

variant	t65 (base 0.573)		t32 (base 0.610)	
	success	Δ	success	Δ
full recipe	0.781	—	0.684	—
– dimension-independent anchor (vector-norm ρ)	0.662	−0.119	0.655	−0.029
– dead zone ($\rho=0$)	0.549	−0.232	0.601	−0.083
– doubled reward step ($\eta_Q=0.5$)	0.769	−0.012	0.615	−0.069
– 3-step returns ($n=1$)	0.782	+0.001	0.512	−0.172
wider trust region ($\delta_{\text{tot}}=0.25$)	0.874	+0.093	0.689	+0.005
stronger reward step ($\eta_Q=2.0$)	0.802	+0.021	0.729	+0.045

drifting base (+0.9). Swapping only the actor update helps on three of the four: CAST on the identical base and budget beats FM + DICE-RL by +5.6 (t32), +6.9 (t65) and +1.6 (t75), reaching 0.738 and 0.878 on the first two, which is +4.8 and +9.1 over that base.

The exception is instructive. On t53 the flow-matching base already scores 0.917 and neither update improves it (−1.8 for CAST, −0.2 for DICE-RL). More generally, the never-below-base behaviour CAST shows on all eight drifting-base tasks does not carry over here: on t53 and t75 it ends 2.0 and 1.2 points below where it started. Fine-tuning a base that is already near its ceiling is a different problem from lifting a weak one, and our trust region is calibrated for the latter.

5.8 Robustness, and what the bound actually does

The remaining knobs are task-dependent, and they track the critic rather than the task’s difficulty. On t32, whose critic carries a strong action-gradient ($\|\nabla_a Q\| \approx 0.55$), 3-step returns are worth 17 points and a doubled reward step 4.5, while widening the movement bound alone does nothing: this task wants more reward signal, not more room to move. On t65, whose gradient is an order of magnitude weaker (≈ 0.035), the pattern inverts — returns and step size are within noise, but widening the trust region is worth +9.3 points. We report this as a regime difference, not a tuned optimum: it is one seed, and t65 carries a ± 9 -point spread, so the direction is trustworthy and the magnitude is not.

Is the gap just a better step size? We tested this directly on t65 by sweeping each method’s own step-size knob under the identical protocol. The backpropagated actor’s learning rate over a $10\times$ range gives 0.550, 0.548 and 0.578 — every setting at or below the untuned base (0.573). Our reward step over an $8\times$ range gives 0.775, 0.817, 0.908 and 0.750 for $\eta_Q \in \{1, 2, 4, 8\}$ — every setting 18–33 points above that same base. The two bands do not overlap at any step size either method was given. A larger step does not turn the backpropagated update into the field update. (Both sweeps are single-seed, so the ordering within each band should not be read as a tuned optimum; the band separation, at 18 points minimum, is well outside seed noise.)

How often does the bound actually bind? The clip only ever reduces a displacement, so whenever a particle’s proposed motion is already shorter than δ_{tot} the update is exactly the unbounded one. In the Section 3.3 setting at our operating point, the clip is active on 17.3% of sampled actions and the median proposed displacement is only $0.12\delta_{\text{tot}}$. Tightening the cap to 0.05 raises the binding rate to 26.3%; loosening it to 1.0 drops it to 4.8%.

These numbers say what kind of object the trust region is. It is not a step-size controller: for most proposals the critic’s own magnitude still sets the distance, exactly as in the unconstrained update. What the constraint governs is the tail — the minority of proposals where an imperfect critic asks for a displacement far outside anything its data supports, and which, left alone, are enough to end the run (Table 5, first row). That is why our claim is about where the critic’s proposal is mediated, not about how far the policy travels

on average. It is also why we do not adjudicate between the two implementations of the constraint: both truncate the same tail.

[TODO: Remaining: Figure 5 panels; robomimic dispersion retention (action dispersion and best-of- N before and after fine-tuning against the backprop control — does the field update preserve the multimodality the base was trained for?).]

6 Conclusion

We showed that the natural update primitives for RL fine-tuning of one-step generative policies are, respectively, unavailable without modifying the policy and unsafe when the critic is imperfect, and that the policy class’s own pretraining primitive — field-target regression — provides the missing update rule. CAST expresses reward and anchoring as composable velocity fields over action particles, and recovers the backpropagated direction in the small-step limit. Its working component is the action-space trust region: the critic sets the direction, while a bound — a pointwise clip on each particle’s displacement, a restoring pull outside a radius of the frozen base, or both — sets how far any single update may carry the policy.

The experiments locate the advantage precisely. Given the same base, critic, data and budget, the field-target form extracts far more than backpropagating that critic: +55 points on robomimic square, and +9.4 versus +0.9 on average across the eight hardest LIBERO-90 tasks, where the backpropagated actor falls below its base on the two highest-headroom tasks and CAST improves on every one. The same ordering holds when the base is swapped for the stronger flow-matching policy, so the effect is attributable to the update rule rather than to the starting point. Against a critic-free policy-gradient baseline at matched interaction budget, CAST is at parity (+10.1 versus +9.5 per-task mean on LIBERO; 0.878 versus 0.883 on the flow-matching base’s one high-headroom task) — while requiring none of the noise injection that makes likelihood ratios definable for a deterministic one-step generator in the first place. Ablations confirm the mechanism rather than the hyperparameters, and they constrain what we may claim. Replacing the dead-zone with an always-on penalty ($\rho=0$, exactly a behaviour-cloning term) falls back to base level (0.549 and 0.601 against bases of 0.573 and 0.610), so the constraint must be shaped in action space rather than added to the loss. Switching the two guards on and off independently is decisive in one direction only: removing both collapses success to exactly 0.000 on both tasks, while either alone recovers essentially all of the gain — indeed removing the anchor and keeping only the step bound is among our strongest settings (0.803 on t65). We therefore claim the necessity of an action-space constraint, and not the superiority of the particular one we ship.

Limitations. Three specific results run against our thesis and we state them plainly. On t81 the backpropagated actor ends above CAST (0.653 versus 0.635); on two flow-matching tasks the un-finetuned base beats every fine-tuner including ours (0.917 on t53, 0.620 on t75); and on robomimic can the flow-matching pipeline edges us on two of three seeds. Single-seed variation is also substantial on LIBERO — repeating the frozen recipe on t65 with a second seed moves the result by 8.9 points — so per-task deltas smaller than roughly ten points, including our η_Q and δ_{tot} sweeps, indicate direction rather than magnitude. The claims that survive this variation are the ones separated by margins far larger than it: the +9.4 versus +0.9 mean gap, the non-overlapping step-size bands, and the below-base collapse when the critic’s gradient is deleted.

The gains track how much a critic actually knows. When a single critic is shared across all eight LIBERO tasks its action-gradient collapses and no critic-based update — ours or the state of the art’s — improves on its base; when a base is already near its ceiling, CAST can end slightly below it (−2.0 and −1.2 on two flow-matching tasks), since our trust region is calibrated for lifting weak bases rather than for refining strong ones. Our critic-trust diagnosis is also post hoc: choosing the reward-field resolution online from critic-trust signals is the natural next step, as is testing whether the reward step should adapt per task. Finally, all results are state-based simulation on a single embodiment.

We should be precise about which part of the story generalizes, because two claims are easy to conflate. The update rule is base-agnostic: it consumes a frozen noise-to-action map, a residual head and a critic, and never touches how the base draws its action — which is why it transfers unchanged to a flow-matching base, where it also improves on the state-of-the-art fine-tuner. What does not generalize is the economy of the construction. A drifting model is pretrained by field-target regression, so on that base the fine-tuner reuses machinery the policy class already has; a flow-matching model is pretrained by conditional velocity regression, so there the same update is bolted on rather than inherited (and costs one base forward pass per particle per integration step). The principle we can defend is therefore narrow: what keeps a critic-driven update on the data is a restoring force with a dead zone, not a limit on step size. The distinction is not cosmetic. A step limit constrains speed, and a drift assembled from permitted small steps still arrives; in our isolation the limit alone leaves the policy at zero success. A behaviour-cloning penalty restores, but does so everywhere, including the region where the critic is reliable, and must therefore trade far-field containment against near-field progress with a single coefficient; it also fails alone. The dead zone separates those jobs.

We deliberately claim no more than this. We have no principled rule for choosing ρ — it is tuned, not derived — and the argument for why the dead zone should exist is an argument about matching the regularizer’s spatial profile to the critic’s reliability, not a bound. Deriving ρ from a measure of local critic reliability is the obvious next question, and we leave it open.

Reproducibility statement

[TODO: Code release (training framework + official-harness implant), exact configs (family-shared recipe), seeds, eval protocols (300-ep robomimic; 100×3 powered LIBERO), compute (single L40S/A100 per run, 8–20h), toy experiment script. All numbers in Table 2 are averages, never best-of.]

References

- Abbas Abdolmaleki, Jost Tobias Springenberg, Yuval Tassa, Remi Munos, Nicolas Heess, and Martin Riedmiller. Maximum a posteriori policy optimisation. In International Conference on Learning Representations (ICLR), 2018.
- Kevin Black, Michael Janner, Yilun Du, Ilya Kostrikov, and Sergey Levine. Training diffusion models with reinforcement learning. In International Conference on Learning Representations (ICLR), 2024. arXiv:2305.13301.
- Wonhyeok Choi, Shutong Ding, Minwoo Choi, Jungwan Woo, Kyumin Hwang, Jaeyeul Kim, Ye Shi, and Sunghoon Im. A review of online diffusion policy rl algorithms for scalable robotic control, 2026. URL <https://arxiv.org/abs/2601.06133>.
- Mingyang Deng, He Li, Tianhong Li, Yilun Du, and Kaiming He. Generative modeling via drifting, 2026. URL <https://arxiv.org/abs/2602.04770>.
- Ying Fan, Olivia Watkins, Yuqing Du, Hao Liu, Moonkyung Ryu, Craig Boutilier, Pieter Abbeel, Mohammad Ghavamzadeh, Kangwook Lee, and Kimin Lee. DPOK: Reinforcement learning for fine-tuning text-to-image diffusion models. In Advances in Neural Information Processing Systems (NeurIPS), 2023.
- Scott Fujimoto, Herke van Hoof, and David Meger. Addressing function approximation error in actor-critic methods. In International Conference on Machine Learning (ICML), 2018.
- Scott Fujimoto, David Meger, and Doina Precup. Off-policy deep reinforcement learning without exploration. In International Conference on Machine Learning (ICML), 2019.
- Tuomas Haarnoja, Haoran Tang, Pieter Abbeel, and Sergey Levine. Reinforcement learning with deep energy-based policies. In International Conference on Machine Learning (ICML), 2017.

- Philippe Hansen-Estruch, Ilya Kostrikov, Michael Janner, Jakub Grudzien Kuba, and Sergey Levine. IDQL: Implicit Q-learning as an actor-critic method with diffusion policies. arXiv preprint arXiv:2304.10573, 2023.
- Ayush Jain, Norio Kosaka, Xinhui Li, Kyung-Min Kim, Erdem Biyik, and Joseph J. Lim. Mitigating suboptimality of deterministic policy gradients in complex q-functions, 2025. URL <https://arxiv.org/abs/2410.11833>.
- Dengyang Jiang, Xin Jin, Dongyang Liu, Zanyi Wang, Mingzhe Zheng, Ruoyi Du, Xiangpeng Yang, Qilong Wu, Zhen Li, Peng Gao, Harry Yang, and Steven Hoi. D-opsd: On-policy self-distillation for continuously tuning step-distilled diffusion models, 2026. URL <https://arxiv.org/abs/2605.05204>.
- Dmitry Kalashnikov, Alex Irpan, Peter Pastor, Julian Ibarz, Alexander Herzog, Eric Jang, Deirdre Quillen, Ethan Holly, Mrinal Kalakrishnan, Vincent Vanhoucke, and Sergey Levine. QT-Opt: Scalable deep reinforcement learning for vision-based robotic manipulation. In Conference on Robot Learning (CoRL), 2018.
- Juil Koo, Mingue Park, Jiwon Choi, Yunhong Min, and Minhyuk Sung. Drifting field policy: A one-step generative policy via wasserstein gradient flow, 2026. URL <https://arxiv.org/abs/2605.07727>.
- Aviral Kumar, Aurick Zhou, George Tucker, and Sergey Levine. Conservative Q-learning for offline reinforcement learning. In Advances in Neural Information Processing Systems (NeurIPS), 2020.
- Haozhan Li, Yuxin Zuo, Jiale Yu, Yuhao Zhang, Zhaohui Yang, Kaiyan Zhang, Xuekai Zhu, Yuchen Zhang, Tianxing Chen, Ganqu Cui, Dehui Wang, Dingxiang Luo, Yuchen Fan, Youbang Sun, Jia Zeng, Jiangmiao Pang, Shanghang Zhang, Yu Wang, Yao Mu, Bowen Zhou, and Ning Ding. Simplevla-rl: Scaling vla training via reinforcement learning, 2025. URL <https://arxiv.org/abs/2509.09674>.
- Zechu Li, Rickmer Krohn, Tao Chen, Anurag Ajay, Pulkit Agrawal, and Georgia Chalvatzaki. Learning multimodal behaviors from scratch with diffusion policy gradient, 2024. URL <https://arxiv.org/abs/2406.00681>.
- Yang Liu, Prajit Ramachandran, Qiang Liu, and Jian Peng. Stein variational policy gradient. arXiv preprint arXiv:1704.02399, 2017.
- Guozheng Ma, Lu Li, Haoyu Wang, Zixuan Liu, Pierre-Luc Bacon, and Dacheng Tao. What makes value learning efficient in residual reinforcement learning?, 2026. URL <https://arxiv.org/abs/2602.10539>.
- Max Sobol Mark, Tian Gao, Georgia Gabriela Sampaio, Mohan Kumar Srirama, Archit Sharma, Chelsea Finn, and Aviral Kumar. Policy agnostic rl: Offline rl and online rl fine-tuning of any class and backbone, 2024. URL <https://arxiv.org/abs/2412.06685>.
- David McAllister, Songwei Ge, Brent Yi, Chung Min Kim, Ethan Weber, Hongsuk Choi, Haiwen Feng, and Angjoo Kanazawa. Flow matching policy gradients, 2025. URL <https://arxiv.org/abs/2507.21053>.
- Mitsuhiko Nakamoto, Yuexiang Zhai, Anikait Singh, Max Sobol Mark, Yi Ma, Chelsea Finn, Aviral Kumar, and Sergey Levine. Cal-QL: Calibrated offline RL pre-training for efficient online fine-tuning. In Advances in Neural Information Processing Systems (NeurIPS), 2023.
- Seohong Park, Qiyang Li, and Sergey Levine. Flow Q-learning. arXiv preprint arXiv:2502.02538, 2025.
- Xue Bin Peng, Aviral Kumar, Grace Zhang, and Sergey Levine. Advantage-weighted regression: Simple and scalable off-policy reinforcement learning. arXiv preprint arXiv:1910.00177, 2019.

- Michael Psenka, Alejandro Escontrela, Pieter Abbeel, and Yi Ma. Learning a diffusion model policy from rewards via q-score matching, 2025. URL <https://arxiv.org/abs/2312.11752>.
- Allen Z. Ren, Justin Lidard, Lars L. Ankile, Anthony Simeonov, Pulkit Agrawal, Anirudha Majumdar, Benjamin Burchfiel, Hongkai Dai, and Max Simchowitz. Diffusion policy optimization. In International Conference on Learning Representations (ICLR), 2025. arXiv:2409.00588.
- Idan Shenfeld, Jyothishh Pari, and Pulkit Agrawal. RL’s razor: Why online reinforcement learning forgets less, 2025. URL <https://arxiv.org/abs/2509.04259>.
- Zhanyi Sun and Shuran Song. From prior to pro: Efficient skill mastery via distribution contractive rl finetuning, 2026. URL <https://arxiv.org/abs/2603.10263>.
- Ziyu Wang, Alexander Novikov, Konrad Zolna, Josh Merel, Jost Tobias Springenberg, Scott Reed, Bobak Shahriari, Noah Siegel, Caglar Gulcehre, Nicolas Heess, and Nando de Freitas. Critic regularized regression. In Advances in Neural Information Processing Systems (NeurIPS), 2020.
- Zhaorui Yang, Tianyu Pang, Haozhe Feng, Han Wang, Wei Chen, Minfeng Zhu, and Qian Liu. Self-distillation bridges distribution gap in language model fine-tuning. arXiv preprint arXiv:2402.13669, 2024.
- Tonghe Zhang, Chao Yu, Sichang Su, and Yu Wang. Reinflow: Fine-tuning flow matching policy with online reinforcement learning, 2026. URL <https://arxiv.org/abs/2505.22094>.
- Zhiyuan Zhou, Andy Peng, Charles Xu, Qiyang Li, Tobias Springenberg, Kevin Frans, and Sergey Levine. Test-time gradient guidance of flow policies in reinforcement learning, 2026. URL <https://arxiv.org/abs/2606.11087>.

A Additional results

[TODO: Full seed tables, training curves, hard-8 per-task breakdown.]

B Hyperparameters

[TODO: Family-shared recipe table; official DICE-RL hypers; sweep grids.]

C The kernel transport operator

[TODO: Full form of $\mathcal{V}$: global distance scale, per-radius softmax-softmax affinity, contested-attractor weighting, self-exclusion, per-radius normalization on/off (restoring vs. constant-pull), radii $\{0.02, 0.05, 0.2\}$. Port from `drift_field.py` docstrings.]

D Toy two-regime construction

[TODO: Details of the hard-critic/easy-critic toy; failure of compass updates.]